Online mistake bounds for Gaussian neural networks and priors preserving their function rankings

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Abstract

A small neural implementation can give polynomial learning bounds even when its Gaussian prior probability is extremely small. This leaves a question for online learning under distribution shift: can the learner make only polynomially many mistakes over its entire lifetime, rather than merely attain small test error on one distribution?

We give two results, keeping their different priors explicit. First, the usual Gaussian tanh posterior, with global output-margin support and exact label conditioning, has polynomial total mistakes under arbitrary adaptive switching among finitely many fixed input environments. For environments uniform on arbitrary subsets of the Boolean cube, the bound is independent of learner width and polynomial in target complexity and input dimension at fixed depth. It covers sufficiently small variance at each width, every fixed above-mean-field exponent, and mean field. The proof permits arbitrary probability ratios among surviving neural functions. A small posterior tail is charged only on first appearances, so an infinite stream incurs no horizon-times-tail penalty.

Second, we change the function prior while preserving its entire ordering and all exact ties. Averaging weights $1/[r(r+1)]$ across tied ranks gives a fixed prior whose target negative log probability is at most twice the logarithm of its inclusive rank, plus a constant. Existing neural compression and mass bounds make this logarithmic rank polynomial in target complexity, uniformly in width. Ordinary conditioning under the modified prior therefore gives polynomial total mistakes on every nonanticipating adaptive input sequence. This prior is realizable on the same neural parameter space, but is generally not an independent Gaussian. We also give a near-minimum-energy sampling alternative that does not need exact ties. No arbitrary-order polynomial theorem for the unchanged large-width Gaussian learner is claimed.

1 The distinction between ordering and probability ratios

Consider online prediction of a fixed Boolean function: each input is presented, a label is predicted, and the true label is then revealed. All observations are retained. A useful simplicity principle would bound the total number of mistakes by a polynomial in the complexity of a target implementation, even when the queried input region changes.

Fresh uniform selection among smallest consistent circuits has such a bound on arbitrary input orders: there are at most s gate-count levels below a target of size s, and each level has a logarithmic candidate elimination budget [5]. Real-valued neural energies do not supply that argument. They can have many distinct levels below the target's energy, and equal leading Gaussian rate does not imply equal function probabilities. Exact neural symmetries do create some ties, but neither

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polynomially many levels nor uniform function probabilities within every tied energy level follows from the usual neural compression theorem.

The distinction is visible in a finite example. On points $x_1, \dots, x_m$, let f be constantly positive and let g_i be positive except at x_i. Give g_i energy $1 + i\eta/(m+1)$ and f energy $1 + \eta$, where $\eta > 0$ is arbitrarily small. Exact minimum-energy prediction makes m mistakes on $x_1, \dots, x_m$, despite the narrow energy interval and logarithmic class size $\log(m+1)$. This is an abstract obstruction to an inference from energy counting alone, not a construction of these energies in the Gaussian neural model.

Our first result retains the Gaussian prior and places explicit conditions on how inputs arrive. Our second retains the same functions, likelihood, and prior ranking, but changes the probability ratios. It then permits arbitrary input orders. All arguments are statistical; we do not assert computationally efficient inference or rank evaluation.

2 Neural model and available structural estimates

Let $d \geq 1$, $L \geq 2$, and let N be the common hidden width. Inputs are $x \in \{-1, 1\}^d$, augmented by a constant coordinate. There are $L - 1$ hidden layers with coordinatewise tanh activation and a linear readout:

$$h^1(x) = \tanh(W^1(1, x)), \qquad h^j(x) = \tanh(W^j h^{j-1}(x)), \qquad z_w(x) = v^\mathsf{T} h^{L-1}(x).$$

There are no additional internal biases. The weighted energy is

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{j=2}^{L-1} \|W^j\|_F^2 + N\|v\|_2^2.$$

For a Boolean function g define its strict global-margin event and minimum energy by

$$U_{N,g} = \{w : g(x) z_w(x) > 1/2 \text{ for every } x\}, \qquad \kappa_N(g) = \inf_{w \in U_{N,g}} \kappa_N(w).$$

Let $\mathcal{H}_N$ contain precisely the functions with nonempty $U_{N,g}$. It is a finite set of distinct functions. Each such event is open and has positive mass under every nondegenerate Gaussian prior. Let $U_N = \bigcup_{g \in \mathcal{H}_N} U_{N,g}$ be the label-independent global absolute-margin event.

In the strong-prior parameterization, the energy coordinates are independent centered Gaussians of variance σ^2. Thus individual weight variances are $\sigma^2 N/d$ in the first layer, σ^2 internally, and σ^2/N in the readout. The finite-variance family sets $\sigma^2 = N^{-\gamma}$, giving variances $N^{1-\gamma}/d, N^{-\gamma}, N^{-\gamma-1}$. Mean field is $\gamma = 1$. Write μ for the raw Gaussian and

$$\pi(g) = \frac{\mu(U_{N,g})}{\mu(U_N)}$$

for its induced function prior after global-margin conditioning. The learning posterior further conditions on every revealed label exactly. In particular, repeated inputs have zero posterior error.

The target complexity $C = C_L(f)$ is the least hidden width of a same-depth tanh implementation with signed margin greater than $1/2$ everywhere and every coefficient bounded in magnitude by that width. We assume $C < \infty$ and allow every $N \geq C$. All polynomial claims below hold at fixed depth; for the above-mean-field family the exponent $\gamma > 1$ is also fixed. Their degrees need not be uniform as γ decreases to one. The mean-field conclusion is supplied by a separate estimate. Natural logarithms are used, and $0 < \delta < 1$ denotes a confidence level.

Lemma 1 (Imported neural structural bounds and their budgets)**.** *Fix depth $L \geq 2$, let $C = C_L(f) < \infty$, and suppose $N \geq C$. All function priors in this lemma use the global output margin restriction before training. There are functions $F_L(d, J)$, $A_L(d, C) \geq 1$, $T_L(d, B)$, and $G_L(d, B)$ with polynomial majorants at fixed L such that the following statements hold. The first is the norm count*

$$\log \#\{g \in \mathcal{H}_N : \kappa_N(g) \leq J\} \leq F_L(d, J) \quad (J \geq 1), \qquad \kappa_N(f) \leq B_L(d, C) := d(d+1)C^2 + LC^4.$$

For the specified finite-variance family, the raw target mass satisfies

$$\mu_{N,\gamma}(U_{N,f}) \geq \exp\{-A_L(d, C)N^\gamma/2\} \qquad (\gamma \geq 1).$$

These are the compression, cloning, and target-cylinder bounds of [1, 2, 3], recorded with explicit scalar formulas in [4].

For strong prior, fix any $0 < \beta < 1$ and any admissible finite N. There is $\sigma_0 > 0$ such that for every $0 < \sigma < \sigma_0$, the class

$$\mathcal{A} = \{g \in \mathcal{H}_N : \kappa_N(g) \leq \kappa_N(f)\}$$

contains f and satisfies

$$\log |\mathcal{A}| \leq h := F_L(d, B_L(d, C)), \qquad \pi(\mathcal{A}^c) \leq \beta\pi(f).$$

The threshold may depend on N, d, L, f, β and the activation; no width-uniform variance threshold is asserted. One threshold suffices for all subsequent noiseless histories and the entire input stream.

For every fixed $\gamma > 1$, put $A = A_L(d, C)$ and

$$B_\beta = 4A + 8\log(L/\beta), \qquad a_\beta = [8L(\sqrt{B_\beta} + 1)^{L-2}]^{-1},$$

$$T_\beta = \left\lceil \max\left\{C, [8(d+1)]^{1/(\gamma-1)}, (8/a_\beta)^{2/(\gamma-1)}\right\}\right\rceil, \quad h_\beta = F_L(d, (L+3)B_\beta), \quad J_\beta = AT_\beta^\gamma/2.$$

At $N \geq T_\beta$, the class $\mathcal{A} = \{g : \kappa_N(g) \leq (L+3)B_\beta\}$ contains f and satisfies $\log |\mathcal{A}| \leq h_\beta$ and $\pi(\mathcal{A}^c) \leq \beta\pi(f)$. At $C \leq N < T_\beta$, instead $-\log \pi(f) \leq J_\beta$. Both budgets are polynomially bounded in $d, C, \log(1/\beta)$ at fixed L, γ; their degrees may depend on γ.

For mean-field tanh, the functions T_L, G_L have the following structural meaning: for $B \geq 4$ and $N \geq T_L(d, B)$, the raw total mass of full-margin functions outside a finite class of log cardinality at most $G_L(d, B)$ is at most $Le^{-NB/8}$. For $\gamma = 1$, use the same A, B_β and set

$$T_\beta = \max\{C, T_L(d, B_\beta)\}, \qquad h_\beta = G_L(d, B_\beta), \qquad J_\beta = AT_\beta/2.$$

At $N \geq T_\beta$ there is a finite class $\mathcal{A}$ containing f with $\log |\mathcal{A}| \leq h_\beta$ and $\pi(\mathcal{A}^c) \leq \beta\pi(f)$; at $C \leq N < T_\beta$, $-\log \pi(f) \leq J_\beta$. These budgets are polynomially bounded in $d, C, \log(1/\beta)$ at fixed L.

Proof. The strong-prior assertion follows from $2\sigma^2 \log \mu_{N,\sigma}(U_{N,g}) \to -\kappa_N(g)$ and finiteness of $\mathcal{H}_N$: the total mass of functions with energy strictly above the target's is $o(\pi(f))$. All energy ties stay in $\mathcal{A}$; no assertion about equal weights is needed [1, 4]. For $\gamma > 1$, the spectral truncation estimate gives raw exceptional mass at most $L\exp(-B_\beta N^\gamma/8)$ under the displayed width conditions [2]. Dividing by the raw target mass gives at most

$$L\exp\{-(B_\beta/8 - A/2)N^\gamma\} = L(\beta/L)^{N^\gamma} \leq \beta.$$

The mean-field finite-description theorem gives the identical calculation with exceptional mass $L\exp(-NB_\beta/8)$ and width threshold $T_L(d, B_\beta)$ [3]. In either case the resulting class must contain f, since otherwise its relative outside mass is at least one. At smaller widths, $-\log \pi(f) \leq -\log \mu(U_{N,f}) \leq AN^\gamma/2$. The polynomial bounds and explicit quantities are also collected in [4]. The classes and tail estimates are fixed before the input process; exact conditioning therefore bounds posterior mass outside each such class by β after every realizable history. □

Corollary 2 (Gaussian function ranks)**.** *For any prior covered by Lemma 1, let*

$$R_f = \#\{g \in \mathcal{H}_N : \pi(g) \geq \pi(f)\}.$$

Then $\log R_f \leq H_\beta$*, where* $H_\beta = h$ *in the strong-prior case and* $H_\beta = \max\{h_\beta, J_\beta\}$ *in either finite-variance case. In particular, choosing* $\beta = 1/2$ *gives a bound polynomial in* d, C *at the fixed parameters of the regime, uniformly for every* $N \geq C$*. This rank is unchanged if raw global-margin masses replace the conditioned function probabilities. The scalar* H_β *bounds both proof branches; it does not assert a common relevant class in the smaller-width branch.*

Proof. In a relevant-class branch, every $g \notin \mathcal{A}$ has $\pi(g) \leq \pi(\mathcal{A}^c) \leq \beta\pi(f) < \pi(f)$. Thus all functions counted by R_f, including every exact tie, belong to $\mathcal{A}$, and $\log R_f \leq \log |\mathcal{A}|$. In a direct-mass branch, $R_f\pi(f) \leq 1$, so $\log R_f \leq -\log \pi(f) \leq J_\beta$. Global-margin conditioning divides all raw function masses by one common normalizer and therefore preserves the ranking and its ties. □

When using strong-prior predictive limits, one takes limits of the posterior laws after conditioning on each realizable history. This does not require conditioning a limiting prior, which may assign the target zero mass.

3 Adaptive switching among fixed sources

The number of energy or probability levels causes no difficulty when fresh inputs retain sufficient randomness within each of finitely many fixed sources. The source can change after every observation. The following result extends the finite-class environment arguments of [5] to a posterior with a uniformly controlled tail. Its tail cost depends on the number of distinct possible inputs, rather than on the duration of the stream.

Let $\mathcal{X}$ be a finite domain of size m, with fixed target f. After the first $t-1$ labels, let Π_{t-1} be a distribution on functions that is supported on functions agreeing with all those labels and is determined before the next input is drawn. Suppose a fixed finite set A contains f and satisfies

$$\log |A| \leq h, \qquad \Pi_{t-1}(A^c) \leq \beta \quad \text{after every realizable history.} \tag{1}$$

For an exact consistency posterior with prior π, the sufficient condition

$$\pi(A^c) \leq \beta\pi(f), \qquad \pi(f) > 0, \tag{2}$$

implies (1): the posterior denominator is at least the mass of the always-consistent target. The hypothesis is thus uniform over rare histories as well as typical ones.

Fix distributions $D_1, \ldots, D_K$ on $\mathcal{X}$, with $K \geq 1$. At each round a scheduler chooses a source using the full observed past, then obtains a fresh independent input from that source. Its choice may also use fresh randomness, but not the unrevealed source samples or the current private predictor draw. The learner need not observe the source identity. Predictable mixtures of the fixed sources are included by regarding the source choice as latent. Choose $0 < \varepsilon_i \leq 1$ such that

$$D_i(g \neq f) \in \{0\} \cup [\varepsilon_i, 1] \qquad (g \in A). \tag{3}$$

One may use the smallest positive disagreement probability, or any lower bound on it. If there is no positive disagreement, any such ε_i is valid.

At the actual input X_t, write

$$q_t = \Pi_{t-1}\{g : g(X_t) \neq f(X_t)\}, \qquad S_\infty = \sum_{t \geq 1} q_t.$$

For fresh sampling from Π_{t-1}, let M_∞ be the total mistakes. The draw can occur after X_t is revealed, provided its conditional law is still Π_{t-1}.

Theorem 3 (Posterior learning under interleaved iid sources)**.** *Under (1) and (3), arbitrary adaptive switching among the fixed sources satisfies*

$$\mathbb{E}S_\infty = \mathbb{E}M_\infty \le \sum_{i=1}^{K} \left[1 + (h+1)\left(1 + \log\frac{1}{\varepsilon_i}\right)\right] + \beta m. \tag{4}$$

For $0 < \delta < 1$, define

$$J_i = 1 + \left\lceil \log_2 \frac{1}{\varepsilon_i} \right\rceil, \qquad \Lambda = \sum_{i=1}^{K} J_i,$$
$$a_r = h + \log\frac{r\Lambda}{\delta}, \qquad B_r = (\Lambda + K)a_r + 2K \quad (r = 2, 3).$$

Then

$$\mathbb{P}\left\{S_\infty > (e-1)B_2 + \log\frac{2}{\delta} + \beta m\right\} \le \delta, \tag{5}$$
$$\mathbb{P}\left\{M_\infty > (e-1)(B_3 + \beta m) + 2\log\frac{3}{\delta}\right\} \le \delta. \tag{6}$$

These are separate probability statements. They hold over the entire possibly infinite horizon and therefore also at every stopping time.

Proof. Split each realized-query error probability as

$$q_t = q_t^{\text{in}} + q_t^{\text{out}}, \qquad q_t^{\text{out}} = \Pi_{t-1}\{g \notin A : g(X_t) \ne f(X_t)\}.$$

Every retained function is correct on a previously observed input. Thus, for the actual realized queries,

$$q_t^{\text{out}} \le \beta\, \mathbf{1}\{X_t \notin \{X_1, \dots, X_{t-1}\}\}, \qquad \sum_{t\ge 1} q_t^{\text{out}} \le \beta m. \tag{7}$$

This is a pathwise inequality and needs no assumption on input order. It is different from a bound on risk averaged over the next source draw.

For the inside contribution, couple source i to its own independent iid stream. Let $R_{i,n}$ be the maximum D_i-risk of a member of A agreeing with the first n labels in that stream, ignoring labels from other sources. A function with risk u survives with probability $(1-u)^n$; therefore, for $n \ge 1$,

$$\mathbb{P}\{R_{i,n} > u\} \le \min\{1, e^{h-nu}\}. \tag{8}$$

At a visit to source i after n of its samples have been seen, all inside-A posterior functions survive that local prefix. Consequently, conditional on the observed past and the chosen source, but *before* the new input is drawn,

$$\mathbb{E}[q_t^{\text{in}} \mid \text{past, source } i] \le R_{i,n}. \tag{9}$$

The same bound holds for the indicator of a sampled mistake from a function in A. Arbitrary relative posterior weights within A are allowed. Other sources' observations can only further reduce the set of possible functions.

The random variable $R_{i,n}$ is either zero or at least ε_i. For $u > 0$,

$$\sum_{n\geq 1} \min\{1, e^{h-nu}\} \leq \int_0^\infty \min\{1, e^{h-tu}\}\, dt = \frac{h+1}{u}.$$

For $0 < u < \varepsilon_i$, use instead $\mathbb{P}(R_{i,n} > u) \leq \min\{1, e^{h-n\varepsilon_i}\}$, by union bounding all positive-risk functions. Integrating the tail and using Tonelli's theorem gives

$$\sum_{n\geq 1} \mathbb{E}R_{i,n} \leq (h+1)\left(1 + \log\frac{1}{\varepsilon_i}\right).$$

The initial local round costs at most one. Each local counter (i,n) is used at most once, irrespective of the adaptive interleaving. Summing (9), adding (7), and using conditional expectation for the fresh sampled predictions proves (4).

We give the high-probability argument with a generic $a > 0$. For $1 \leq j \leq J_i$, set $n_{i,j} = \lceil a2^j \rceil$. Except on an event of probability at most Λe^{h-a}, all prefix inequalities

$$R_{i,n_{i,j}} \leq 2^{-j}$$

hold. Since $2^{-J_i} < \varepsilon_i$, the last one forces risk zero. Monotonicity of $R_{i,n}$ supplies a deterministic envelope: use one before $n_{i,1}$, use 2^{-j} from $n_{i,j}$ until $n_{i,j+1} - 1$, and use zero from n_{i,J_i} onwards. Its sum over all local counts for source i is at most

$$2a + 1 + \sum_{j=1}^{J_i - 1} (a + 2^{-j}) \leq (J_i + 1)a + 2.$$

Stop the inside process before using any prefix that violates this applicable envelope. Whether the current prefix violates it is known from the observed local history before drawing the next input. Until stopping, the sum of conditional mean bounds is at most $B = (\Lambda + K)a + 2K$ on every path. For either the stopped fractional variable q_t^{in} or the stopped inside-mistake indicator, $0 \leq Y_t \leq 1$ and

$$\mathbb{E}[e^{Y_t} \mid \text{past, chosen source}] \leq 1 + (e-1)\mathbb{E}[Y_t \mid \text{past, chosen source}].$$

The exponential-supermartingale argument therefore bounds its total by $(e-1)B + \log(1/\eta)$ apart from probability η. This holds uniformly in time. On the simultaneous good-prefix event, stopping never occurs. Taking $a = a_2$, $\eta = \delta/2$, and adding the pathwise outside sum proves (5).

For sampled mistakes use $a = a_3$ and $\eta = \delta/3$ for the inside process. For the outside process a different conditioning stage is appropriate: condition on the past, the chosen source, and the *realized current input*, immediately before the private function draw. The outside-mistake indicator Z_t^{out} then has conditional mean q_t^{out}, and

$$\mathbb{E}\left[e^{Z_t^{\text{out}} - (e-1)q_t^{\text{out}}} \mid \text{past, source, current input}\right] \leq 1.$$

Taking the further expectation over the source and input makes the product a nonnegative supermartingale at the ends of complete rounds. By (7), its compensator is bounded pathwise by $(e-1)\beta m$. Thus outside mistakes total at most $(e-1)\beta m + \log(3/\delta)$, apart from probability $\delta/3$. A union bound over prefix failure, the inside tail, and the outside tail proves (6). Infinite sums follow by monotone limits, or directly from the time-uniform supermartingale bounds. □

Pointwise, the number of posterior-majority mistakes is at most $2S_\infty$, and the number of failures to assign the correct label posterior probability strictly greater than $2/3$ is at most $3S_\infty$. Thus (4) gives these expectation bounds with factors two and three, respectively; (5) gives their high-probability counterparts. Majority prediction may depend on the query; it is controlled through this posterior-risk comparison, not by assuming the majority function itself belongs to A.

Finite pools. Fix K finite pools of sizes $m_i \geq 1$, independently uniformly permute each, and adaptively interleave their unrevealed entries. Repeated inputs within or between pools are allowed. The random-pool prefix argument of [5], applied to the inside posterior, gives

$$\mathbb{E}M \leq K + 2(h+1)\sum_{i=1}^{K} H_{m_i-1} + \beta m, \qquad H_0 = 0. \tag{10}$$

Indeed, after n entries of pool i, the conditional inside risk is at most $m_i R_{i,n}/(m_i - n)$, and $\sum_{n=1}^{m_i-1} m_i/[n(m_i-n)] = 2H_{m_i-1}$. The outside term is still (7), regardless of duplicate entries. For pools with at most 2^d entries, this is $O(K(h+1)(d+1)) + \beta 2^d$. The same stopping and tail split gives a sampled high-probability bound $(e-1)(B+\beta m) + 2\log(3/\delta)$, using $a = h + \log(3\sum_i m_i/\delta)$ and $B = K + 2a\sum_i H_{m_i-1}$. For cumulative posterior risk use the corresponding two-event allocation, as in (5).

Corollary 4 (Structured shifts for the original Gaussian NNBayes learners)**.** *Use any of the strong-prior, fixed-$\gamma > 1$, or mean-field tanh regimes of Lemma 1, with the same target and depth qualifications. Let $m = 2^d$, $\beta = 1/m$, and let H_β majorize both the lemma's relevant-class logarithmic count h_β and its direct prior-cost budget J_β where that alternative is used. For $K \geq 1$ fixed sources uniform on arbitrary nonempty subsets of the cube,*

$$\mathbb{E}M_\infty \leq K\big[1 + (H_\beta + 1)(1 + d\log 2)\big] + 1. \tag{11}$$

With

$$a = H_\beta + \log\frac{3K(d+1)}{\delta}, \qquad B = K(d+2)a + 2K,$$

one also has

$$\mathbb{P}\left\{M_\infty > (e-1)(B+1) + 2\log\frac{3}{\delta}\right\} \leq \delta. \tag{12}$$

Thus at fixed depth, and fixed γ where applicable, these are polynomial total-mistake bounds in $d, C, K, \log(1/\delta)$, independently of learner width. The scheduler may switch indefinitely and arbitrarily abruptly among the fixed sources. Corresponding majority and confidence bounds follow from the cumulative posterior risk. Fixed finite pools with at most 2^d entries each have the same polynomial conclusion.

Proof. In the relevant-class branch of Lemma 1, the prior-ratio condition gives (1) after every history. Every positive disagreement probability on a uniform subset of the cube is at least 2^{-d}, so use $\varepsilon_i = 2^{-d}$ for every source. Then $J_i = d+1$, $\Lambda = K(d+1)$, $h \leq H_\beta$, and $\beta m = 1$. Theorem 3 gives the displayed bounds.

In the direct-prior-cost branch, no common finite class is needed or asserted. Exact Bayesian telescoping gives $S_\infty \leq -\log\pi(f) \leq J_\beta \leq H_\beta$ for every input sequence, and posterior sampling has tail $(e-1)H_\beta + \log(1/\delta)$. These are bounded by the displayed budgets. The same branch distinction gives the posterior-risk claim. For finite pools take the maximum of the direct budget and the relevant-class budget in (10); no finite-class assertion is made in the direct branch. The imported budgets are polynomial after substituting $\log(1/\beta) = d\log 2$. □

For strong prior the variance is chosen sufficiently small at each fixed width before the stream begins, with the original possibility that its threshold depends on width. One such choice covers all histories and all subsequent switches. Any simultaneous limit of the correctly conditioned predictive posteriors as the variance tends to zero at that fixed width also satisfies the result: each limiting posterior is supported on the target-energy sublevel, giving $\beta = 0$. This refers to limits *after*

conditioning on each history; conditioning a limiting prior that has assigned zero mass to the target is not the argument.

The sources themselves must be fixed in advance. A polynomial number of sources with arbitrary switching is a substantive extension of the uniform-stream theorem, but does not cover every deterministic order: encoding an arbitrary cube order as singleton sources uses $K = 2^d$.

4 Rank flattening of the function prior

All logarithms in this section are natural. Let $\mathcal{H}$ be a finite set of distinct Boolean functions, let $M = |\mathcal{H}| \geq 1$, and let π be a strictly positive prior on $\mathcal{H}$. For $f \in \mathcal{H}$, define its strict and inclusive ranks by

$$a_f = |\{g : \pi(g) > \pi(f)\}|, \qquad b_f = |\{g : \pi(g) \geq \pi(f)\}|.$$

Define the *rank-flattened prior*

$$Z_M = \frac{M}{M+1}, \qquad q_\pi(f) = \frac{1}{Z_M(a_f+1)(b_f+1)}. \tag{13}$$

This transformation is performed once, before observing any labels. Afterwards, q_π is updated by ordinary conditioning on consistency.

Proposition 5 (Normalization, preferences, and target cost)**.** *The law q_π is a probability distribution and preserves every strict ordering and every tie of π:*

$$\pi(f) > \pi(g) \iff q_\pi(f) > q_\pi(g), \qquad \pi(f) = \pi(g) \iff q_\pi(f) = q_\pi(g).$$

In particular, the two consistency posteriors have the same sets of maximum-a-posteriori functions on every history. Writing $R_f = b_f$,

$$\log \frac{1}{q_\pi(f)} = \log Z_M + \log(a_f+1) + \log(b_f+1) \leq \log(R_f(R_f+1)) \leq 2\log(R_f+1). \tag{14}$$

Proof. Order functions by decreasing π-mass. A tie block occupies consecutive ranks $a+1, \ldots, b$. Its mean weight under $w_r = 1/[r(r+1)]$ is

$$\frac{1}{b-a}\sum_{r=a+1}^{b} \frac{1}{r(r+1)} = \frac{1}{b-a}\left(\frac{1}{a+1} - \frac{1}{b+1}\right) = \frac{1}{(a+1)(b+1)}.$$

Thus the unnormalized masses in (13) sum to $\sum_{r=1}^{M} w_r = Z_M$. Since w_r is strictly decreasing, its averages over successive disjoint rank blocks strictly decrease. This proves the ordering and tie assertions. Conditioning multiplies all surviving masses by a common positive constant, so it preserves the maximizing set. Finally $a_f + 1 \leq b_f = R_f$ and $Z_M < 1$ give (14). □

Theorem 6 (Arbitrary-order prediction after rank flattening)**.** *Fix a target $f_* \in \mathcal{H}$. At round t, an input x_t is chosen, the learner predicts, and the label $f_*(x_t)$ is revealed. The inputs may depend on the whole preceding history, including past randomized predictions, but are chosen before the learner's fresh randomization at the current round. Let q_t be the consistency posterior of the fixed prior q_π, and set*

$$p_t = q_t\{g : g(x_t) = f_*(x_t)\}, \qquad B_* = \log \frac{1}{q_\pi(f_*)}.$$

Uniformly over such input-selection rules, including infinite streams, the following hold.

1. *Fresh posterior sampling has expected total mistakes at most B_*, which is bounded by (14).*

2. *Posterior majority, with any tie rule, has at most $B_*/\log 2$ mistakes.*

3. *The number of rounds with $p_t \le 2/3$ is at most $B_*/\log(3/2)$.*

For fresh posterior sampling, with probability at least $1-\delta$, the total number of mistakes is also at most $(e-1)B_* + \log(1/\delta)$, for every $0 < \delta < 1$.

Proof. Let V_t be the functions consistent with the first $t-1$ labels. Because $f_* \in V_t$, all posterior denominators are positive. On every realized path and at every finite horizon T,

$$\prod_{t=1}^{T} p_t = q_\pi(V_{T+1}) \ge q_\pi(f_*), \qquad \sum_{t=1}^{T} -\log p_t \le B_*. \tag{15}$$

The conditional mistake probability of a fresh posterior draw is $u_t = 1 - p_t \le -\log p_t$. Taking expectations and then applying monotone convergence proves the first assertion, including the infinite horizon. A majority mistake implies $p_t \le 1/2$, and a round with $p_t \le 2/3$ contributes at least $\log(3/2)$ to (15). This proves the next two assertions pathwise.

For the last assertion, let I_t be the sampled mistake indicator. Conditionally on the history and x_t,

$$\mathbb{E}[e^{I_t}] = 1 + (e-1)u_t \le e^{(e-1)u_t}.$$

Consequently $\exp(\sum_{t\le T} I_t - (e-1)\sum_{t\le T} u_t)$ is a nonnegative supermartingale starting at one. Since $\sum_{t\le T} u_t \le B_*$ pathwise, Markov's inequality gives the stated bound at each finite horizon. Letting the horizon increase proves the same bound for the total number of mistakes. □

The theorem concerns posterior sampling and posterior majority, not posterior MAP. In fact Proposition 5 shows that rank flattening leaves the MAP prediction rule unchanged if its tie rule is unchanged. Nor does the theorem concern repeatedly rank flattening a changing posterior: its proof uses one fixed predata prior.

Corollary 7 (Arbitrary-order neural learning with preserved function rankings)**.** *Use any Gaussian regime and eligible width in Lemma 1, and transform its actual induced function prior by (13). With the rank budget $H_{1/2}$ from Corollary 2, the online guarantees of Theorem 6 hold with*

$$B_* \le 2H_{1/2} + \log 2.$$

This is polynomial in d, C at fixed depth, and fixed $\gamma > 1$ where applicable, independently of learner width. It includes the positive mean-field variance. In the strong-prior case it holds for every sufficiently small positive variance at each fixed width, with the variance threshold independent of the stream and its length.

Proof. Corollary 2 gives $\log R_f \le H_{1/2}$. Use (14) and $\log[R(R+1)] \le 2\log R + \log 2$, then apply Theorem 6. Neither the construction of the transformed prior nor its posterior updates require knowing the target or its complexity. That complexity is used only to prove the bound. □

An alternative ranks directly by increasing minimum energy $\kappa_N(g)$, treating equal energies as ties. Its inclusive target rank is at most the energy-sublevel count from Lemma 1. The same construction then gives an arbitrary-order budget $2F_L(d, B_L(d, C)) + \log 2$ at every $N \ge C$, without a variance parameter. It preserves energy ordering, rather than exact finite-variance Gaussian probability ordering. No limiting equality of prefactors is assumed.

Realization as a parameter prior. Let μ^U be the original parameter law conditioned on the global margin event, and let $g(w) \in \mathcal{H}$ be its induced function, so $\mu^U\{w : g(w) = f\} = \pi(f)$. The probability law

$$\frac{d\nu}{d\mu^U}(w) = \frac{q_\pi(g(w))}{\pi(g(w))} \tag{16}$$

has induced function law q_π. Indeed, its integral over the event $\{g(w) = f\}$ is $q_\pi(f)$. It preserves the conditional distribution of parameters within each function event. This is a global reweighting of function masses; it generally destroys Gaussianity and coordinate independence. It is neither ordinary Gaussian temperature adjustment nor an efficient sampling claim. Raising a function mass to a power and raising a parameter density to that power are different operations, since integration and taking powers do not generally commute.

An alternative that samples near the minimum energy. Exact real-valued energies need not be tied for a related construction to work. Let $E : \mathcal{H} \to [0, \infty)$ and $H_j = \{g : E(g) \le 2^j\}$ for $j \ge 0$. On each history choose the least j for which H_j has a consistent function, and sample uniformly over its distinct consistent functions. Retain every past label when the budget increases. If the least surviving energy is $m \ge 1$, every sampled energy is at most $2m$; for $m < 1$, it is at most one.

Corollary 8 (Geometric energy bands)**.** *For the preceding learner, put* $J_* = \lceil \log_2 \max\{1, E(f_*)\} \rceil$. *Then*

$$\mathbb{E}[\textit{total mistakes}] \le J_* + \sum_{j=0}^{J_*} \log \max\{1, |H_j|\}.$$

In particular, if $\log |H_j| \le A 2^{jp}$ *whenever* H_j *is nonempty, with* $A > 0$ *and* $p > 0$, *this bound is at most*

$$J_* + A \frac{2^{p(J_*+1)} - 1}{2^p - 1},$$

and hence polynomial in the target energy for fixed p. *The neural energy and entropy bounds of Lemma 1 give a width-independent polynomial at fixed depth.*

Proof. The selected budget never decreases and never exceeds J_*. During a fixed-budget phase, let n_t be its number of consistent functions. Except on a round that empties the phase, the error probability obeys $1 - n_{t+1}/n_t \le \log(n_t/n_{t+1})$, so these costs telescope to at most $\log |H_j|$. Each emptied phase costs at most one additional mistake; there are at most J_* such phases because H_{J_*} contains the target. Sum the bounds and, for an infinite stream, use monotone convergence. □

A fixed Bayesian alternative mixes the uniform laws on nonempty H_j with weights proportional to 2^{-j-1}. Its target negative log-mass is at most $\log |H_{J_*}| + (J_* + 1) \log 2$, so the potential argument (15) immediately gives the corresponding online bounds. Both energy constructions introduce ties between different energies in one geometric bin; rank flattening preserves every original strict probability preference.

5 What changes and what remains unresolved

The structured-environment theorem concerns the original Gaussian posterior. It never replaces unequal posterior weights by uniform ones. The sources are fixed before the process, while switches or mixture weights can depend on the full observed past. Allowing an arbitrary new source at no cost would invalidate the conclusion: encoding every adversarial query as its own singleton source

makes their number as large as the domain. The theorem therefore supplies a concrete class of distribution shifts, rather than a claim that most imaginable sequences are benign.

The prior transformation concerns a different learner. It preserves which function is preferred to which, but changes how strongly it is preferred. This is enough to improve the general online bound because the logarithm of the number of functions preferred to a small target is polynomial, even when the original target probability is extremely small. Since the maximizing functions do not change, the improvement is a theorem about posterior sampling and posterior majority, not about choosing one posterior mode.

Function-level flattening also differs from increasing the variance of the original independent Gaussian weights. Function masses integrate parameter density over different regions of parameter space; changing a scalar temperature does not generally remove their volume differences. No theorem here identifies either function-level construction with an independent-coordinate weight prior. Nor does counting low-energy functions alone establish their Gaussian mass ratios.

For the unchanged Gaussian learner at unrestricted large width, a polynomial total-mistake bound for every adaptive input order remains open in the fixed-depth setting considered here. Its elementary bound $-\log \pi(f)$ can grow with width or diverge as variance decreases; this fact alone is not a lower bound on actual mistakes. The geometric bands and rank-based prior are precise positive alternatives, not a resolution of that remaining question. The global-margin support condition and fixed-depth quantifiers remain part of all neural applications in this paper.

Verification. The new online transfers and function-prior constructions have conventional proofs and independent AI-agent mathematical reviews. They are not Lean formalizations. Existing formal verification of the imported neural learning results does not automatically certify these new consequences.

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